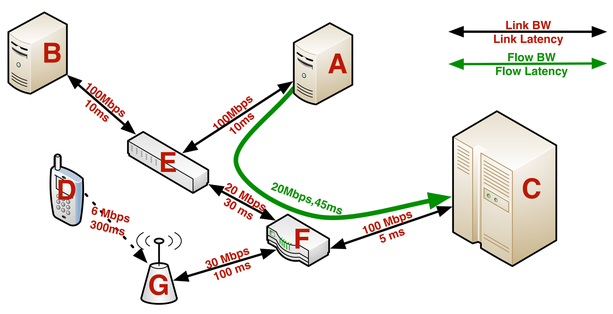
User-Space Network Emulation for the SPLAY Platform



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Network Emulation

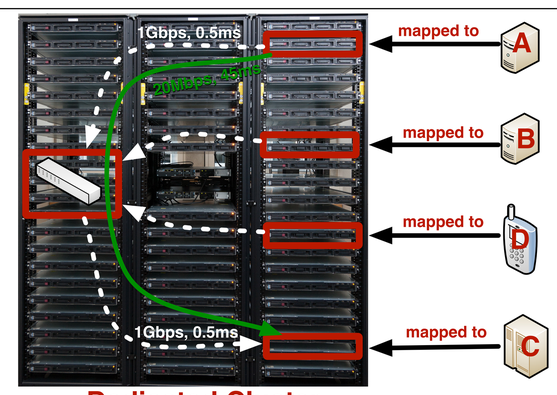


Network Topology to Emulate

```
<topology>
    <vertices>
        <vertex int_idx="1" role="virtnode" int_vn="A" />
    <vertex int_idx="2" role="virtnode" int_vn="B" />
    <vertex int_idx="3" role="virtnode" int_vn="C" />
    <vertex int_idx="4" role="virtnode" int_vn="D" />
    <vertex int_idx="5" role="gateway" int_vn="E" />
    <vertex int_idx="6" role="gateway" int_vn="F" />
    <vertex int_idx="7" role="gateway" int_vn="G" />
    </vertices>
     <edges>
         <edge src="A" dest="E" delayms="10" mbps="100" />
         <edge src="B" dest="E" delayms="10" mbps="100" />
         <edge src="E" dest="F" delayms="30" mbps="20" />
         <edge src="D" dest="G" delayms="300" mbps=6" />
         <edge src="G" dest="F" delayms="100" mbps="30" />
         <edge src="F" dest="C" delayms="5" mbps="100" />
    </edges>
</topology>
```

Topology Representation in DSL

Apply link shaping



Dedicated Cluster

Context and Problem

- Testbeds to evaluate distributed protocols
- Require privileged access
- Tedious setup, (expensive) dedicated HW
- Many under-used machines available



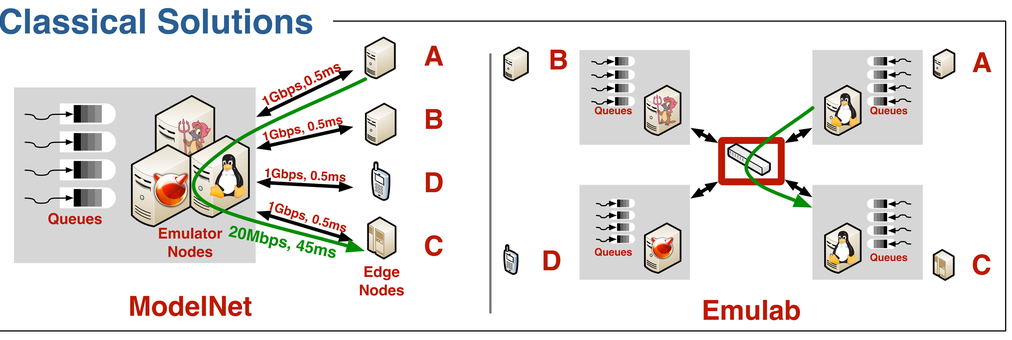
Objectives

- Multiple network topologies at the same time
- Cross-platform execution
- No kernel modifications
- Reach performances of dedicated hardware

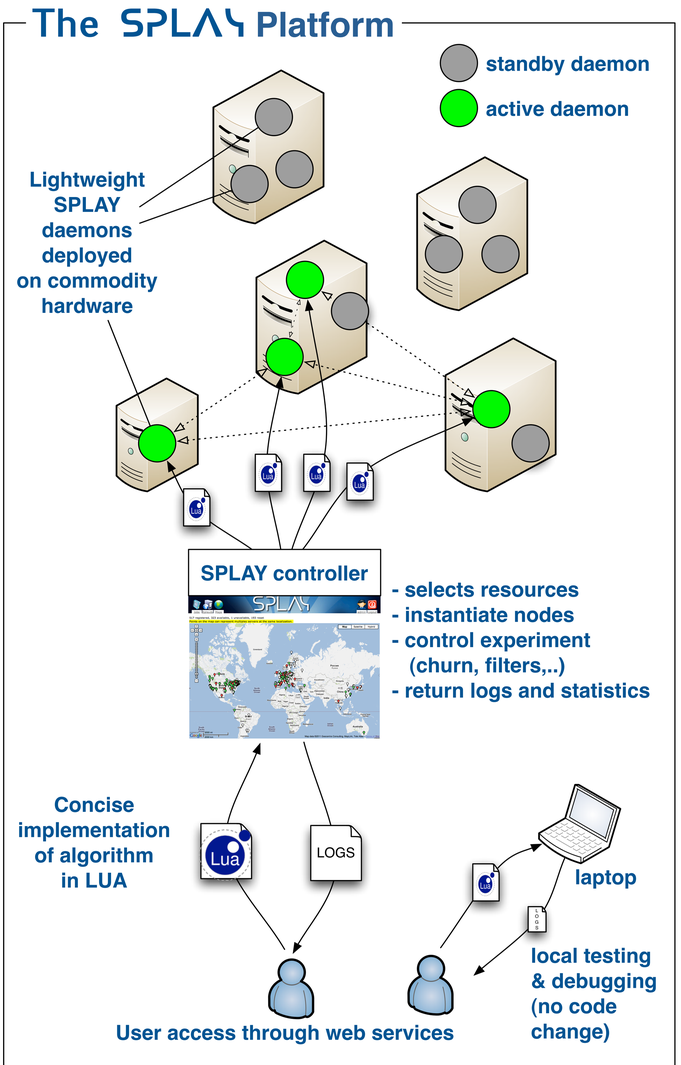
Challenges

- Emulation of internal nodes and links:
  - Congestion, BW, delays
- No dedicate traffic shaping node
- React to fast detection of traffic changes

Classical Solutions

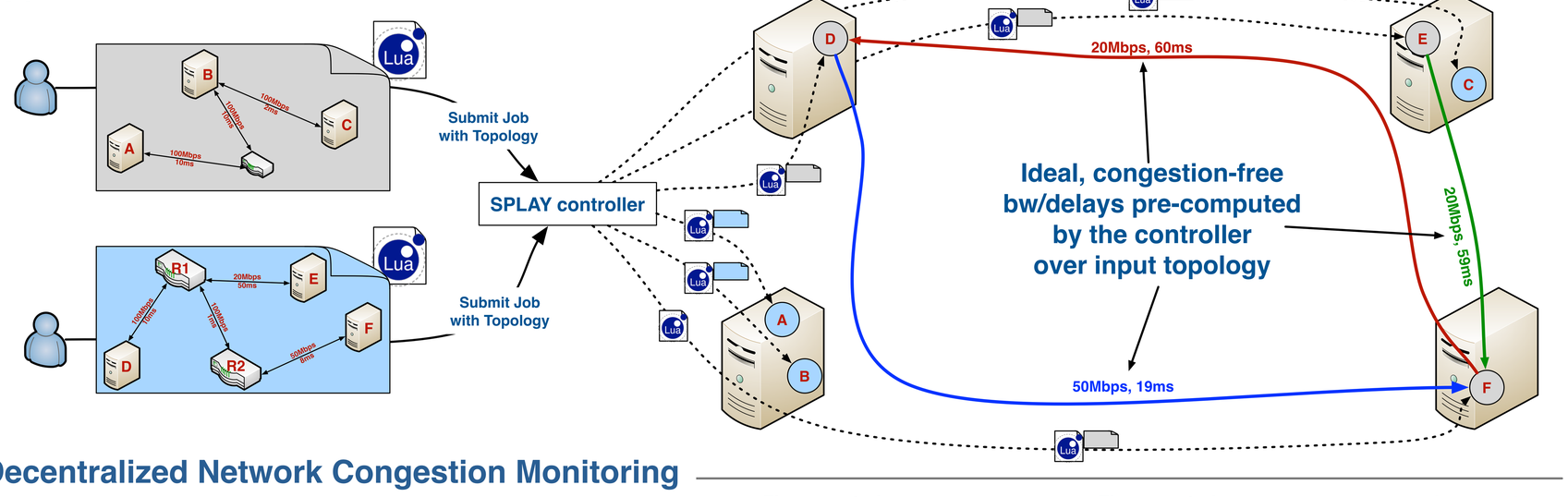


The SPLAY Platform



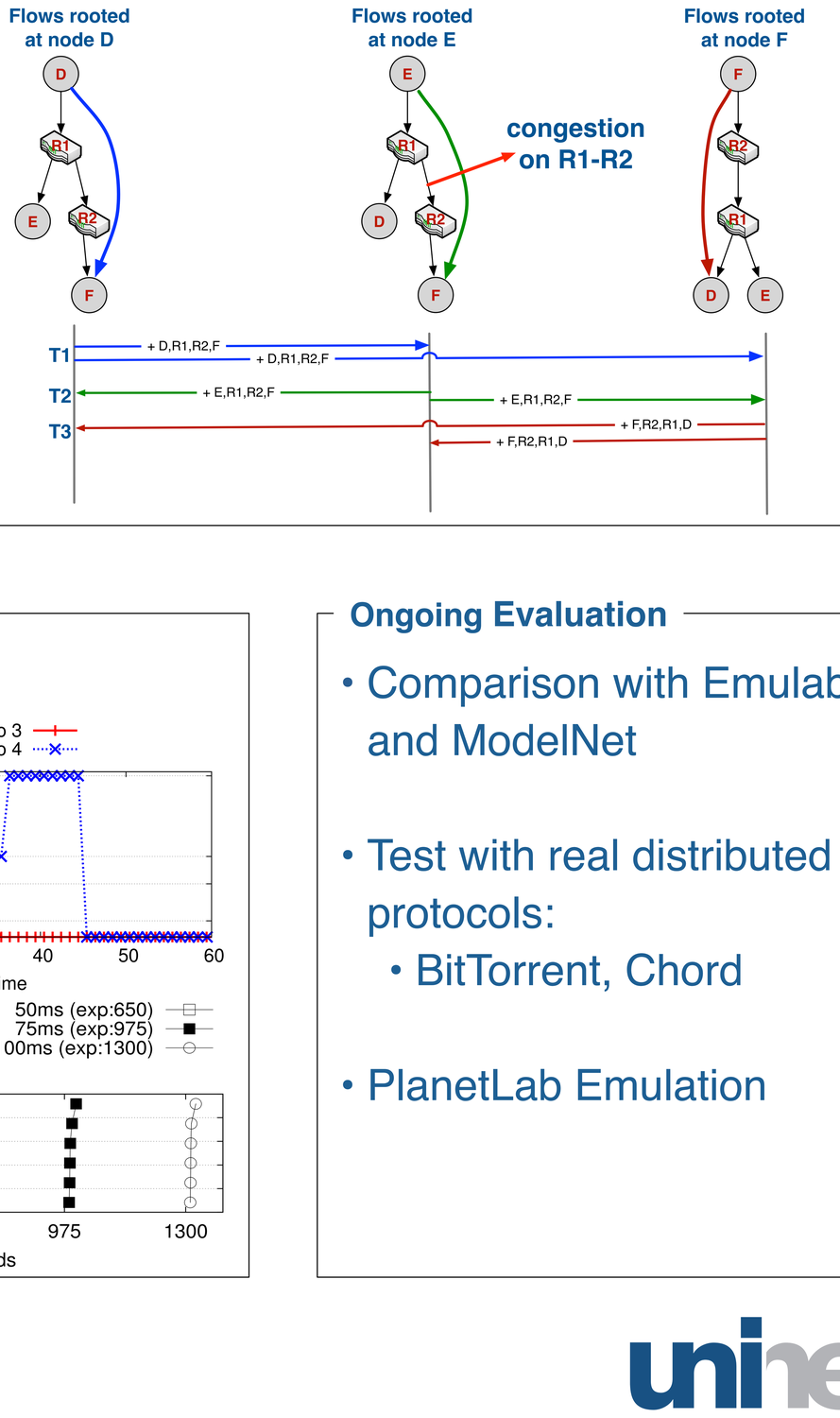
www.splay-project.org

Application-Level Network Emulation for SPLAY



Decentralized Network Congestion Monitoring

- Lightweight diffusion protocol to notify nodes about current data flows
- Nodes adjust BW rates according to the competing flows on bottleneck links
- At T2, the congestion on bottleneck link R1-R2 triggers BW throttling

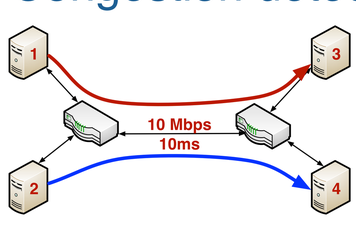


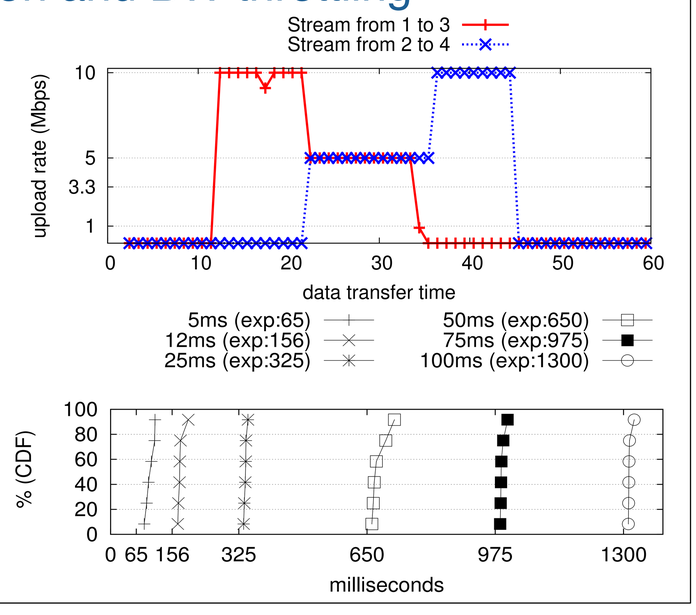
Implementation

- Pure Lua
- Congestion monitoring using UDP/Multicast (for LANs)
- BW shaping using Hierarchical Token Bucket
- Support for ModelNet's and Emulab's topology DSL

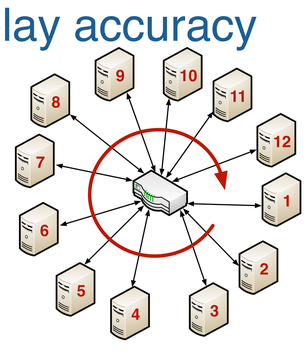
Evaluation

- Congestion detection and BW throttling





- Delay accuracy



Ongoing Evaluation

- Comparison with Emulab and ModelNet
- Test with real distributed protocols:
  - BitTorrent, Chord
- PlanetLab Emulation

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